

MOHIT KUMAR

Pune, Maharashtra, India

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Education

Army Institute of Technology, Pune

Bachelor of Engineering in Computer Science(CGPA: 8.57)

Aug 2024 - Aug 2028

Pune, Maharashtra

Technical Skills

Languages: Java, TypeScript, Python, C++, SQL

Frameworks: Spring Boot, Node.js, FastAPI, React, Next.js

Technologies: REST APIs, WebSockets, Caching, Redis, Kafka, FAISS

Developer Tools: Linux, Git, Docker, Kubernetes, AWS, CI/CD, Prometheus

Projects

ChatFlow | *Spring Boot, WebSockets, Redis, PostgreSQL* | GitHub

- Built a real-time messaging backend using Spring Boot and WebSockets, integrating Redis Pub/Sub to synchronize message broadcasting across horizontally scaled instances.
- Reduced chat history retrieval latency by **~60%** by implementing a Redis read-through cache.
- Added custom circuit breakers, JWT-based authentication, and Redis-backed token-bucket rate limiting, validated to handle **10,000** concurrent connections via distributed load testing using Apache JMeter.

Cyclotrack | *Node.js, React, WebSockets, Redis* | GitHub

- Built a real-time group cycling tracker using for the college Cycling Club with MERN stack and WebSockets.
- Reduced server RAM consumption from 512MB to a stable **100MB** (**~80%**), resolving OOM crashes by implementing **50ms** payload debouncing for continuous GPS pings and utilizing volatile Socket.io events.
- Maintained a smooth **60 FPS** client-side mapping UI using state management and component memoization, preventing expensive DOM re-renders.

Distributed URL Shortener | *Java, Spring Boot, Redis, PostgreSQL, Kafka* | GitHub

- Implemented a Redis read-through cache and Bloom filters to eliminate database bottlenecking and cache penetration.
- Architected a Base62 short-code generator utilizing a token-range allocation strategy, eliminating synchronous database collision checks and supporting **10,000+** creations/sec.
- Decoupled click analytics using asynchronous message queues, improving read throughput by **~70%** while capturing rich GeoIP and User-Agent metrics.

Achievements & Leadership

- **Specialist** on Codeforces (max rating 1513) | Profile
- Solved **500+** questions across platforms (Leetcode, Cf) | Profile
- Winner of **2x** hackathons, selected from more than **500** participating teams.
- Joint Secretary, **Competitive Programming** Club and PR Cell, **AIT Pune**.
- Organized 2 large-scale coding competitions (**Codeft**) with **3000+** footfall.

Certifications

Operating Systems

NPTEL (IIT Madras)

2025

Database Management Systems

NPTEL (IIT Bombay)

2025